

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

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Chemistry — Class IX | Paper: English Medium

ANSWER KEY (Version 2)

Syllabus: Chapter 1 — Nature of Chemistry | Chapter 2 — Matter | Chapter 3 — Atomic Structure | Chapter 4 — Periodic Table | Chapter 5 — Chemical Bonding | Chapter 6 — Stoichiometry | Chapter 7 — Electrochemistry | Chapter 8 — Energetics | Chapter 9 — Chemical Equilibrium

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	Answer	Explanation
1	Which branch of chemistry studies physical and chemical changes in living organisms?	B 1 mk	Biochemistry is defined as the study of physical and chemical changes occurring in living organisms. Medicinal chemistry deals with drug design; analytical chemistry identifies substances; nuclear chemistry studies nuclear changes.
2	Which state of matter is electrically conductive and found in the sun and lightning?	C 1 mk	Plasma is an ionized gas that conducts electricity, found in the sun, lightning, and stars. BEC forms near absolute zero; liquid crystals are intermediate states; gels are not a standard state of matter.
3	Which was a major defect of Rutherford's atomic model?	B 1 mk	According to classical electrodynamics, an accelerating electron should lose energy and spiral into the nucleus , and should emit a continuous spectrum. Rutherford's model could not explain why electrons remain in stable orbits or why only a line spectrum is emitted.
4	An atom has atomic number 17 and mass number 35. How many neutrons does it have?	C 1 mk	Neutrons = Mass number – Atomic number = 35 – 17 = 18 neutrons . This is the chlorine-35 isotope. The atomic number (17) gives the number of protons; the difference gives neutrons.
5	Which of the following elements has the highest electronegativity on the Pauling scale?	C 1 mk	Fluorine (F = 4.0) has the highest electronegativity on the Pauling scale, as it is the most electronegative element. Electronegativity increases across a period and decreases down a group; F is in Period 2, Group 17.
6	Which halogen exists as a red-brown liquid at room temperature?	C 1 mk	Bromine (Br₂) is a red-brown liquid at room temperature. Fluorine is a pale-yellow gas; chlorine is a yellow-green gas; iodine is a grey-black solid. Physical state of halogens changes from gas → liquid → solid down the group.
7	In a coordinate covalent bond, both bonding electrons are donated by:	C 1 mk	In a coordinate (dative) covalent bond , both electrons come from one atom, called the donor . The receiving atom is called the acceptor. Example: NH ₃ donates a lone pair to H ⁺ to form NH ₄ ⁺ .
8	What is the formula mass of KCl?	B 1 mk	Formula mass of KCl = K + Cl = 39 + 35.5 = 74.5 amu . NaCl = 23 + 35.5 = 58.5 amu (option A, which is a common confusion). KCl is used as a substitute for table salt.
9	How many molecules are present in 0.5 mol of CH ₄ ?	B 1 mk	Number of molecules = moles × Avogadro's number = 0.5 × 6.022 × 10 ²³ = 3.011 × 10²³ molecules. One full mole = 6.022 × 10 ²³ ; half a mole = half that number.
10	In the reaction 2Na + Cl ₂ → 2NaCl, which substance is the	B 1 mk	Cl₂ is the oxidizing agent because it is reduced (gains electrons: Cl ₂ → 2Cl ⁻ , oxidation state goes from 0 to -1). Na is oxidized

	oxidizing agent?		(loses electrons, $0 \rightarrow +1$) and is the reducing agent. The oxidizing agent is reduced in the reaction.
11	A catalyst increases the rate of a chemical reaction by:	B 1 mk	A catalyst lowers the activation energy by providing an alternative reaction pathway with lower energy barrier. It does not change the enthalpy of the reaction or consume itself. Enzymes act as biological catalysts using the same principle.
12	At chemical equilibrium, which of the following is true?	D 1 mk	Chemical equilibrium is dynamic : both the forward and reverse reactions continue simultaneously, but at the same rate. Therefore concentrations of reactants and products remain constant, but the reactions have not stopped.

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q 2(i): Differentiate between industrial chemistry and nuclear chemistry. Give one example of each. 3 mks

Industrial chemistry is the branch of chemistry concerned with the large-scale production of chemical substances for commercial use. 1 mk Example: production of fertilizers such as urea or ammonia by the Haber process.

Nuclear chemistry is the branch that studies changes in the nuclei of atoms, including radioactive decay, fission, and fusion. 1 mk Example: radioactive decay of Carbon-14 to Nitrogen-14, or Uranium-238 to Lead-206. 1 mk

OR

Explain the difference between science, technology, and engineering with one example each.

Science is the systematic study of the natural world through observation and experimentation. Example: studying how atoms bond together. 1 mk

Technology is the application of scientific knowledge to solve practical problems. Example: using knowledge of combustion reactions to design car engines. 1 mk

Engineering is the use of science and mathematics to design, construct, and operate systems and structures. Example: designing a chemical plant to manufacture plastics. 1 mk

Q 2(ii): Compare the macroscopic properties of solids, liquids, and gases in tabular form. 3 mks

Property	Solid	Liquid	Gas
Density	High	High	Low
Compressibility	Not compressible	Moderately compressible	Very compressible
Ability to flow	Cannot flow	Can flow	Can flow

(1 mark per correct row or two correct columns) 3 mks

OR

What are allotropes? Describe the structural differences between graphite and diamond, and why graphite conducts electricity but diamond does not.

Allotropes are different structural forms of the same element in the same physical state. 1 mk

Graphite: Each carbon atom is bonded to 3 others in flat 2D hexagonal layers. The fourth electron per carbon is delocalized between layers, allowing graphite to **conduct electricity**. The layers slide easily, making graphite a lubricant. 1 mk

Diamond: Each carbon atom is bonded to 4 others in a rigid 3D tetrahedral network. All four electrons per carbon are used in bonding, leaving **no free electrons**, so diamond does not conduct electricity. It is the hardest natural substance. 1 mk

Q 2(iii): State any THREE postulates of Bohr's atomic model. How did Bohr explain the emission of light? 3 mks

1. Electrons move in fixed circular orbits (energy levels) around the nucleus without losing energy.
2. Each orbit has a fixed, definite energy; energy is proportional to distance from the nucleus.
3. Electrons can jump from a lower orbit to a higher one by **absorbing energy**, and fall back by **emitting a photon** of light with energy $\Delta E = E_2 - E_1$.

(1 mark per postulate, up to 2 marks; 1 mark for light emission explanation) 3 mks

OR

Define isotopes. Name the three isotopes of hydrogen, give their mass numbers, and identify which is radioactive.

Isotopes are atoms of the same element that have the same atomic number but different mass numbers (different numbers of neutrons). 1 mk

Name	Symbol	Mass Number	Neutrons
Protium	^1H	1	0
Deuterium	^2H	2	1
Tritium	^3H	3	2

Tritium (^3H) is radioactive. 2 mks

Q 2(iv): Write the electronic configuration of Na (Z=11) and Al (Z=13). State their group and period. 3 mks

Sodium (Na, Z=11): $1s^2 2s^2 2p^6 3s^1 \rightarrow$ Group 1, Period 3. 1.5 mks

Aluminium (Al, Z=13): $1s^2 2s^2 2p^6 3s^2 3p^1 \rightarrow$ Group 13, Period 3. 1.5 mks

Both elements are in Period 3 because their outermost electrons are in the $n=3$ shell. Na has 1 valence electron (Group 1); Al has 3 (Group 13).

OR

Explain the trend in ionization energy across a period and down a group.

Across a period (left to right): Ionization energy **increases**. As nuclear charge increases across a period (more protons), electrons are held more tightly. The number of shells stays the same so shielding is approximately constant. It becomes harder to remove an electron. 1.5 mks

Down a group: Ionization energy **decreases**. Going down a group, a new electron shell is added. The increased shielding by inner electrons reduces the effective nuclear attraction on the outermost electron, making it easier to remove. Example: Li = 520, Na = 496, K = 419 kJ/mol. 1.5 mks

Q 2(v): Define ionic bond. Explain how the formula of CaCl_2 is derived from ionic charges. 3 mks

Ionic bond is the strong electrostatic force of attraction between oppositely charged ions (a positively charged metal cation and a negatively charged non-metal anion). It forms when an electropositive metal transfers electrons to an electronegative non-metal. 1 mk

Deriving the formula of CaCl_2 : 2 mks

1. Identify the ions: Ca^{2+} (charge = +2) and Cl^- (charge = -1).
2. The compound must be electrically neutral. To balance +2, we need two Cl^- ions.
3. Formula: CaCl_2 (one calcium ion for every two chloride ions).

OR

Distinguish between polar and non-polar covalent bonds with two examples of each. What determines bond polarity?

Polar covalent bond: Forms between two **different** atoms with different electronegativities. The more electronegative atom pulls the shared electrons closer, creating partial charges (δ^+ and δ^-). Examples: H-Cl (in HCl), H-O (in H_2O). 1.5 mks

Non-polar covalent bond: Forms between **identical** atoms with the same electronegativity. Electrons are shared equally, no partial charges. Examples: H-H (in H_2), O=O (in O_2). 1 mk

Polarity is determined by the **difference in electronegativity** between the two bonded atoms. Greater difference = more polar bond. 0.5 mk

Q 2(vi): Find the empirical formula of a compound: 40% C, 6.7% H, 53.3% O. 3 mks

Step 1 – Moles of each element (assume 100 g): C: $40 / 12 = 3.33$ mol H: $6.7 / 1 = 6.70$ mol O: $53.3 / 16 = 3.33$ mol Step 2 – Divide by smallest (3.33): C: $3.33/3.33 = 1$ H: $6.70/3.33 = 2.01 \approx 2$ O: $3.33/3.33 = 1$ Empirical Formula: CH_2O

Award: 1 mk for moles calculation, 1 mk for ratio, 1 mk for correct formula. 3 mks

Note: CH_2O is the empirical formula of formaldehyde and also of glucose.

OR

Calculate the mass of 2.5 mol of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) and find the number of molecules.

Molar mass of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$: C: $12 \times 12 = 144$ H: $22 \times 1 = 22$ O: $11 \times 16 = 176$ Total M = 342 g/mol Mass = n $\times$ M = $2.5 \times 342 = 855$ g Number of molecules = $2.5 \times 6.022 \times 10^{23} = 1.506 \times 10^{24}$ molecules

1 mk molar mass correct; 1 mk mass = 855 g; 1 mk molecules = 1.506×10^{24}

Q 2(vii): Balance $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$ and identify the reaction type. 3 mks

Unbalanced: $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$ Balance Al: need 4 Al on left (to get $2 \times \text{Al}_2\text{O}_3$ on right) Balance O: $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ Check: Left: 4 Al, 6 O | Right: 4 Al, 6 O ✓ Balanced equation: $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$

1.5 mks balanced equation; 1 mk reaction type: **combination (synthesis) reaction**; also a **redox reaction** (Al is oxidized, O is reduced). 0.5 mk

OR

Calculate moles in 100 g of H_2O and in 5 g of H_2 .

Moles of H_2O : $n = m/M = 100/18 = 5.56$ mol (M of $\text{H}_2\text{O} = 2+16 = 18$ g/mol) Moles of H_2 : $n = m/M = 5/2 = 2.5$ mol (M of $\text{H}_2 = 1+1 = 2$ g/mol)

1.5 mks each calculation correct.

Q 2(viii): State FOUR rules for assigning oxidation states. Find oxidation state of Cr in $K_2Cr_2O_7$.

3 mks

1. Oxidation state of free (uncombined) elements is zero.
2. For simple (monoatomic) ions, oxidation state equals the charge on the ion.
3. In a complex ion, sum of oxidation states equals the charge on the ion.
4. Sum of oxidation states in a neutral compound is zero.

1 mk (any 2 correct rules)

$K_2Cr_2O_7$: K is Group 1 $\rightarrow$ oxidation state = +1; 2 K atoms: total = +2 O has oxidation state = -2; 7 O atoms: total = -14 Let Cr = x; 2 Cr atoms: total = 2x Sum = 0: +2 + 2x + (-14) = 0 $2x - 12 = 0$ $x = +6$
Oxidation state of Cr in $K_2Cr_2O_7$ = +6

2 mks for correct working and answer.

OR

Define oxidizing and reducing agents. In $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$, identify each agent.

Oxidizing agent: A substance that causes another substance to be oxidized by accepting electrons from it. The oxidizing agent is itself reduced in the reaction. **1 mk**

Reducing agent: A substance that causes another substance to be reduced by donating electrons to it. The reducing agent is itself oxidized. **1 mk**

In $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$: Fe in Fe_2O_3 goes from +3 to 0 (reduced), so **Fe_2O_3 is the oxidizing agent**. Carbon in CO goes from +2 to +4 (oxidized), so **CO is the reducing agent**. **1 mk**

Q 2(ix): What is corrosion? Describe TWO methods to prevent it. **3 mks**

Corrosion is the natural electrochemical process in which a metal reacts with oxygen and moisture in the atmosphere, converting refined metals back to more stable metal oxides. The most familiar example is rusting of iron ($Fe_2O_3 \cdot xH_2O$). **1 mk**

Method 1 — Painting: Applying paint or epoxy creates a physical barrier between the metal surface and the environment (oxygen and moisture). It is cheap and easy to apply. **1 mk**

Method 2 — Galvanizing: A clean iron sheet is dipped in a hot zinc chloride bath and then rolled in a zinc bath. The zinc layer sacrificially protects the iron beneath, as zinc is more reactive and corrodes preferentially. **1 mk**

Accept also: alloying (adding Cr to form stainless steel) or electroplating or cathodic protection as valid alternatives.

OR

When acidified $KMnO_4$ is added to a reducing agent, what colour change occurs? Explain. What is the change in oxidation number of Mn?

The solution changes from **purple to colourless**. **1 mk**

Potassium permanganate ($KMnO_4$) is a strong oxidizing agent. When added to a reducing agent (such as $FeSO_4$), the MnO_4^- ions (purple) are reduced to Mn^{2+} ions (colourless), which discharges the purple colour. **1 mk**

The oxidation number of Mn changes from **+7** (in MnO_4^-) to **+2** (in Mn^{2+}), a decrease of 5. This represents reduction. **1 mk**

Q 2(x): Define enthalpy of reaction. State the sign of ΔH for exo/endothermic reactions. Give one example each. 3 mks

Enthalpy of reaction is the amount of heat or thermal energy evolved or absorbed when a chemical reaction takes place. It is denoted by ΔH . 1 mk

Exothermic reaction: ΔH is **negative** (heat released to surroundings). Example: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H^\circ = -393.5 \text{ kJ}$ 1 mk

Endothermic reaction: ΔH is **positive** (heat absorbed from surroundings). Example: $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$ $\Delta H^\circ = +53.8 \text{ kJ}$ 1 mk

OR

Define activation energy. Explain how a catalyst changes it, with reference to the energy pathway diagram.

Activation energy is the minimum amount of energy that colliding molecules must possess, in addition to their average kinetic energy, for a collision to be effective (i.e., to result in a chemical reaction). 1 mk

A **catalyst** provides an alternative reaction pathway with a **lower activation energy**. On the energy pathway diagram, the catalyzed path shows a lower peak (transition state) compared to the uncatalyzed path. 1 mk

Because more molecules now have sufficient energy to react, the rate of reaction increases. Importantly, the catalyst does **not** change the overall enthalpy change (ΔH) of the reaction — only the height of the energy barrier. Enzymes are biological catalysts that work by the same principle. 1 mk

Q 2(xi): What is a reversible reaction? Write the equation for hydrated copper(II) sulphate dehydration. State colour changes. 3 mks

A **reversible reaction** is one in which the products can react with each other to re-form the original reactants. Such reactions proceed in both forward and reverse directions simultaneously under the same conditions. Shown by the symbol $\rightleftharpoons$. 1 mk

Equation: $CuSO_4 \cdot 5H_2O$ (blue solid) $\rightleftharpoons$ $CuSO_4$ (white solid) + $5H_2O$ 1 mk

On **heating**: blue hydrated copper(II) sulphate loses water and becomes **white** anhydrous copper(II) sulphate. On **adding water**: white solid absorbs water and turns **blue** again. 1 mk

OR

State THREE conditions for equilibrium. What is dynamic equilibrium?

Three conditions for equilibrium: 2 mks

1. The concentration of reactants or products remains unchanged.
2. The temperature of the system remains constant.
3. The pressure or volume of the system remains constant.

Dynamic equilibrium means that both the forward and reverse reactions are still occurring at the same rate; the system is not static. The concentrations of reactants and products remain constant because the rates of the two opposing reactions are equal. 1 mk

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q 3(i)(a): Dalton's atomic theory — three postulates and one limitation. 3 mks

Dalton's Atomic Theory (1803):

1. All matter is composed of tiny, indivisible particles called **atoms**.
2. All atoms of a given element are identical in mass and properties; atoms of different elements differ in mass and properties.
3. Atoms combine in simple whole-number ratios to form compounds.

(Award 1 mark per postulate, up to 2 marks) **2 mks**

Limitation: Dalton stated atoms are indivisible, but we now know atoms consist of subatomic particles (protons, neutrons, electrons). His model also could not explain isotopes (atoms of the same element with different masses). **1 mk**

Q 3(i)(b): Protons, neutrons, electrons for element Z=19, A=39. Electronic config, group, period. 2 mks

Protons = Z = 19 Neutrons = A - Z = 39 - 19 = 20 Electrons = 19 (neutral atom) Electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 \rightarrow$ Group 1, Period 4 (This is Potassium, K)

1 mk for protons/neutrons/electrons; **1 mk** for config + group/period.

OR

Q 3(i) OR (a): How is the periodic table organized? Define s-, p-, d-block with examples. 3 mks

The periodic table contains 118 elements arranged in order of increasing **atomic number**. There are 7 periods (horizontal rows) and 18 groups (vertical columns). **1 mk**

s-block: Groups 1 and 2; outermost electrons in s-orbital. Example: Na (Group 1), Mg (Group 2). **1 mk**

p-block: Groups 13-18 (except He); outermost electrons in p-orbital. Example: Al (Group 13), Cl (Group 17). **0.5 mk**

d-block: Transition metals (Groups 3-12); outermost electrons in d-orbital. Example: Fe (iron), Cu (copper). **0.5 mk**

Q 3(i) OR (b): Atomic size trends across a period and down a group. 2 mks

Across a period (left to right): Atomic size **decreases**. Nuclear charge increases (more protons added) but electrons are added to the same shell. The stronger nuclear pull draws electrons closer, shrinking the atom. **1 mk**

Down a group: Atomic size **increases**. Each successive element has an additional electron shell further from the nucleus, increasing the atomic radius despite greater nuclear charge (shielding effect of inner shells). **1 mk**

Q 3(ii)(a): Define mole and Avogadro's number. Calculate mass of 9.05 mol of O₃. 3 mks

Mole: One mole is the amount of a substance that contains as many particles (atoms, molecules, or ions) as there are atoms in exactly 12 g of Carbon-12. 0.5 mk

Avogadro's number: The number of particles in one mole = 6.022×10^{23} . 0.5 mk

Molar mass of O₃: $M = 3 \times 16 = 48 \text{ g/mol}$ Mass = $n \times M = 9.05 \times 48 = 434.4 \text{ g}$

1 mk molar mass = 48; 1 mk mass = 434.4 g (accept 434 g).

Q 3(ii)(b): Balanced net ionic equation for Mg + H₂SO₄. Identify spectator ions. 2 mks

Full equation: $\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$ Full ionic form: $\text{Mg} + 2\text{H}^+ + \text{SO}_4^{2-} \rightarrow \text{Mg}^{2+} + \text{SO}_4^{2-} + \text{H}_2$ Spectator ion: SO_4^{2-} (present on both sides, unchanged) Net ionic equation: $\text{Mg} + 2\text{H}^+ \rightarrow \text{Mg}^{2+} + \text{H}_2$

1 mk net ionic equation; 1 mk SO_4^{2-} identified as spectator ion.

OR

Q 3(ii) OR (a): Empirical vs molecular formula. Find molecular formula of benzene given empirical CH, M = 78 amu. 3 mks

Empirical formula is the simplest whole-number ratio of atoms in a compound. **Molecular formula** gives the actual number of each type of atom in one molecule of the compound. 1 mk

Empirical formula of benzene: CH Empirical formula mass = $12 + 1 = 13 \text{ amu}$ $n = \text{Molecular mass} / \text{Empirical formula mass} = 78 / 13 = 6$ Molecular formula = $(\text{CH})_6 = \text{C}_6\text{H}_6$

1 mk calculation of $n=6$; 1 mk molecular formula C_6H_6 .

Q 3(ii) OR (b): Atoms in 1.25 mol Zn and in 0.2 mol Al. 2 mks

Atoms in 1.25 mol Zn: $= 1.25 \times 6.022 \times 10^{23} = 7.528 \times 10^{23} \text{ atoms} \approx 7.53 \times 10^{23}$ Atoms in 0.2 mol Al: $= 0.2 \times 6.022 \times 10^{23} = 1.2044 \times 10^{23} \text{ atoms}$

1 mk each calculation correct.

Q 3(iii)(a): Define redox reactions in terms of (i) O transfer, (ii) H transfer, (iii) electron transfer. One example each. 3 mks

(i) In terms of oxygen: Oxidation = gain of oxygen; Reduction = loss of oxygen.

Example: $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$. CO gains oxygen (oxidized); Fe_2O_3 loses oxygen (reduced). 1 mk

(ii) In terms of hydrogen: Oxidation = loss of hydrogen; Reduction = gain of hydrogen.

Example: $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$. C_2H_2 loses hydrogen (oxidized); O_2 gains hydrogen (reduced).

1 mk

(iii) In terms of electrons: Oxidation = loss of electrons; Reduction = gain of electrons.

Example: $\text{Cu}^{2+} + \text{Zn} \rightarrow \text{Cu} + \text{Zn}^{2+}$. Zn loses 2 electrons (oxidized); Cu^{2+} gains 2 electrons (reduced). 1 mk

Q 3(iii)(b): Oxidation state of S in H_2SO_4 and N in HNO_3 . 2 mks

H_2SO_4 : $\text{H} = +1$, $\text{O} = -2$ $2(+1) + x + 4(-2) = 0$ $2 + x - 8 = 0$ $x = +6 \rightarrow \text{S in H}_2\text{SO}_4 = +6$ HNO_3 : $\text{H} = +1$, $\text{O} = -2$ $(+1) + x + 3(-2) = 0$ $1 + x - 6 = 0$ $x = +5 \rightarrow \text{N in HNO}_3 = +5$

1 mk each oxidation state with working.

OR

Q 3(iii) OR (a): Describe cathodic protection. Principle and one application. 3 mks

Cathodic protection (also called sacrificial protection) is a method of preventing corrosion in which the metal to be protected is made the cathode and is connected to a more active metal (such as magnesium or aluminium) that acts as the anode. 1 mk

Principle: The more active metal (e.g., Mg) is more reactive than iron and therefore oxidizes preferentially. It sacrifices itself, releasing electrons, while the iron is protected because it receives electrons and is kept reduced (acts as cathode). 1 mk

Application: Large bars of magnesium are attached to the hulls of ships below the waterline. The Mg blocks corrode instead of the iron hull, protecting the ship. Cathodic protection is also used on underground pipelines, oil rigs, and tanks. 1 mk

Q 3(iii) OR (b): $\text{CuO} + \text{Zn} \rightarrow \text{Cu} + \text{ZnO}$ — identify oxidized/reduced elements and agents. 2 mks

$\text{CuO} + \text{Zn} \rightarrow \text{Cu} + \text{ZnO}$ Oxidation states: Cu in CuO = +2 $\rightarrow$ Cu (metal) = 0 (decrease: REDUCED) Zn (metal) = 0 $\rightarrow$ Zn in ZnO = +2 (increase: OXIDIZED) Therefore: Zn is oxidized $\rightarrow$ Zn is the reducing agent Cu^{2+} is reduced $\rightarrow$ CuO is the oxidizing agent

1 mk correct identification of oxidized/reduced; 1 mk correct agents named.

Q 3(iv)(a): Calculate ΔH for $H_2 + Cl_2 \rightarrow 2HCl$ using bond energies. State exo/endothermic. 3 mks

Bond energies: $H-H = 436 \text{ kJ/mol}$, $Cl-Cl = 243 \text{ kJ/mol}$, $H-Cl = 432 \text{ kJ/mol}$ $\Delta H^\circ = (\text{Sum of bond dissociation energies of reactants}) - (\text{Sum of bond formation energies of products})$ Bonds broken (reactants): $1 \times H-H \text{ bond} = 436 \text{ kJ}$ $1 \times Cl-Cl \text{ bond} = 243 \text{ kJ}$ Total energy absorbed = $436 + 243 = 679 \text{ kJ}$ Bonds formed (products): $2 \times H-Cl \text{ bonds} = 2 \times 432 = 864 \text{ kJ}$ Total energy released = 864 kJ $\Delta H^\circ = 679 - 864 = -185 \text{ kJ/mol}$

$\Delta H^\circ = -185 \text{ kJ/mol}$. Since ΔH is negative, the reaction is **exothermic** (heat is released). 3 mks

(1 mk bonds broken; 1 mk bonds formed; 1 mk correct sign and exothermic conclusion)

Q 3(iv)(b): What is aerobic respiration? Word equation. Why is it exothermic? 2 mks

Aerobic respiration is the biochemical process in which glucose is oxidized in the presence of oxygen in living organisms to release energy. 1 mk

Word equation: Glucose + Oxygen $\rightarrow$ Carbon dioxide + Water + Energy 0.5 mk

Aerobic respiration is **exothermic** because energy is released to the biological system (ΔH is negative). The products (CO_2 and H_2O) have lower energy than the reactants (glucose + O_2), so the excess energy is released as heat to maintain body temperature and to power cellular processes. 0.5 mk

OR

Q 3(iv) OR (a): Effect of temperature on cobalt(II) chloride equilibrium. Equation and colour changes. 3 mks

Hydrated cobalt(II) chloride and anhydrous cobalt(II) chloride exist in the following reversible equilibrium:



On heating: Temperature increases, equilibrium shifts to the right (forward reaction is endothermic, absorbs heat to reduce the change). Hydrated cobalt(II) chloride (pink) loses water and becomes anhydrous cobalt(II) chloride (**blue**). 1 mk

On adding water (cooling): The reverse reaction is favoured. The anhydrous blue solid absorbs water and reverts to hydrated cobalt(II) chloride (**pink**). This illustrates that when temperature is increased, equilibrium shifts to favour the endothermic direction to minimise the temperature change. 1 mk

Q 3(iv) OR (b): Define system and surroundings. Give one exothermic and one endothermic thermochemical equation. 2 mks

System: The part of the universe under study in a chemical reaction (e.g., the reactants and products in a test tube). **Surroundings:** Everything else in the universe outside the system (e.g., the test tube, the water bath, and the air around it). 1 mk

Exothermic: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H^\circ = -393.5 \text{ kJ}$ (heat released to surroundings)

Endothermic: $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$ $\Delta H^\circ = +180.5 \text{ kJ}$ (heat absorbed from surroundings) 1 mk